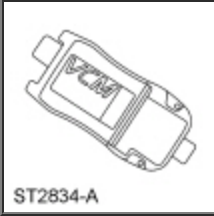


Preliminary Inspection

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
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Material

Item	Specification
Motorcraft® MERCON® LV Automatic Transmission Fluid XT-10-QLVC (US); CXT-10-LV12 (Canada)	MERCON® LV

Check the following items prior to beginning the diagnostic procedures:

Know and Understand the Concern

To correctly diagnose a concern, first understand the customer concern or condition. Customer contact may be required to understand the conditions, including when the concern occurs. For example:

- Hot or cold vehicle temperature
- Hot or cold ambient temperature
- Vehicle driving conditions
- Vehicle loaded/unloaded

After understanding when and how the concern occurs, proceed to Verification of Condition.

Verification of Condition

This section provides information that must be used in determining the actual cause of customer concerns and carrying out the appropriate procedures.

Use the following procedures when verifying customer concerns for the transmission.

Determine Customer Concern

Determine customer concerns relative to vehicle use and dependent driving conditions, paying attention to the following items:

- Hot or cold vehicle operating temperature
- Hot or cold ambient temperature
- Type of terrain
- Vehicle loaded/unloaded
- City/highway driving
- Upshifting
- Downshifting

- Coasting
- Engagement
- Noise/vibration — check for engine rpm, vehicle speed, shift, gear, range or temperature dependencies.

Check Transmission Fluid Level and Condition

Transmission Fluid Level Check

NOTICE: The vehicle should not be driven if the transmission fluid level is low. Internal transmission failure could result.

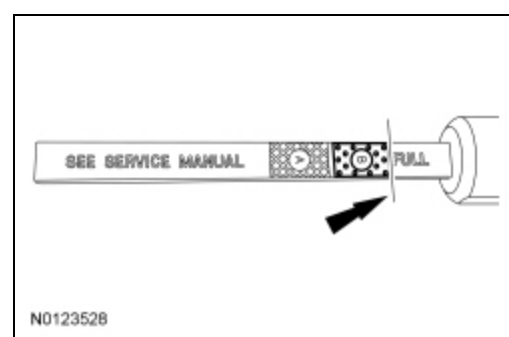
NOTICE: The transmission fluid fill plug is located near the exhaust system. The exhaust may be extremely hot during this procedure.

NOTE: If the vehicle is operated for an extended period at high highway speeds, in city traffic, during hot weather, or while pulling a trailer, allow the transmission fluid to cool down to obtain an accurate reading.

NOTE: If the transmission starts to slip, shifts slowly, or shows signs of transmission fluid leaking, check the transmission fluid level.

This transmission is equipped with an internal thermal bypass located in the case. This internal thermal bypass valve shuts off transmission fluid flowing to the transmission fluid cooler if the transmission fluid temperature falls below normal operating temperature. Refer to Transmission Fluid Level Check.

The transmission fluid level indicator has 2 areas for the fluid level, a crosshatched area (labeled A) and a dotted area (labeled B). Use the dotted (labeled B) area when checking the transmission fluid level. The correct transmission fluid level is at the upper level of the dotted marks on the transmission fluid level indicator.



High Transmission Fluid Level

NOTICE: Transmission fluid level that is too high can cause the transmission fluid to become aerated due to the churning action of the rotating internal parts. This may cause erratic control pressure, foaming, loss of transmission fluid through the vent tube, and possible transmission failure.

If an overfill reading is indicated on the transmission fluid level indicator, remove the excess transmission fluid. Refer to Transmission Fluid Level Check.

Low Transmission Fluid Level

Low transmission fluid level can result in poor transmission engagement, slipping, or failure. This may also indicate a leak in one of the transmission seals or gaskets.

Adding Transmission Fluid

NOTICE: The use of any type of transmission fluid other than specified can result in transmission failure.

NOTICE: The transmission fluid fill plug is located near the exhaust system. The exhaust may be extremely hot during this procedure.

NOTICE: The vehicle should not be driven if the transmission fluid level is low. Internal failure could result.

Add transmission fluid using the refill procedure. Refer to Transmission Fluid Drain and Refill.

Transmission Fluid Condition Check

1. Check the transmission fluid level.
2. Remove the transmission fluid fill plug and transmission fluid level indicator, allowing the transmission fluid to drip onto a facial tissue. Examine the stain.
3. Observe the color and the odor.
4. If evidence of solid material is found, remove the transmission fluid pan for further inspection.
5. If the stain is a foamy pink color, this may indicate coolant in the transmission. Inspect the engine cooling system at this time.
6. If transmission fluid contamination or transmission failure is confirmed by the sediment in the bottom of the transmission fluid pan, disassemble and completely clean the transmission. This includes the torque converter and transmission fluid cooler tubes. Refer to Section 307-02.

Water in Transmission Fluid

To correctly repair an automatic transmission that has had water or coolant introduced into the system, completely disassemble, clean, and replace the following parts:

- All internal and external seals
- All friction material
- Torque converter
- All parts with bonded seals
- All solenoids
- All transmission fluid filters

Prior to installing the transmission, flush and clean the transmission fluid cooler(s) and the transmission fluid cooler tubes and hoses.